Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/05/2026
PI Name:	Rutkove, Seward M.D.		
Project Title:	Resveratrol and Electrical Muscle Stimulation as Countermeasures to Preserve Sensorimotor Function During a 60-Day Head-Down Bedrest Protocol		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	srutkove@bidmc.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-667-8130
Organization Name:	Beth Israel Deaconess Medical Center, Inc./Harvard Medical School		
PI Address 1:	330 Brookline Ave TCC-810		
PI Address 2:			
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/01/2021	**End Date:**	04/30/2028
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2028 per C. Ribeiro/JSC (Ed., 6/15/23) NOTE: End date changed to 1/31/2023 per NSSC information (Ed., 2/7/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Beheshti, Afshin Ph.D. (NASA Ames Research Center) Bouxsein, Mary Ph.D. (Beth Israel Deaconess Medical Center, Inc./Harvard Medical School) Mortreux, Marie Ph.D. (Beth Israel Deaconess Medical Center, Inc.) Aenor, Sawyer (University of California, San Francisco)		
Grant/Contract No.:	80NSSC21K0369		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	New approaches to preserve sensorimotor function during and after periods of prolonged microgravity are needed for planned extended stays in space. These approaches, termed countermeasures, have mainly utilized exercise protocols to date. Here we seek to objectively study two other forms of countermeasures: the nutraceutical resveratrol and electrical muscle stimulation. We will investigate the impact of both of these therapies alone and together in a group of individuals undergoing 60 days of head-down bed rest, a standard method for mimicking microgravity on Earth. We will determine whether these two therapies alone or in combination can help maintain sensorimotor function as compared to individuals receiving placebo or sham therapy. We will also assess relevant physiological pathways via serological analyses as well as any adverse effects related to the therapies. This work is highly significant since it will provide new insights into two novel and relatively easily instituted approaches for maintaining sensorimotor function in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We will identify new approaches for maintaining muscle health that could be applicable to individuals experiencing extended periods of inactivity, such as during severe illnees requiring intensive care unit stays or with injuries precluding them from normal activity, such as with trauma, stroke, or spinal cord injury. Our understanding of the effects of electrical stimulation could have important ramifications in that context.
Task Progress:	We have now completed two of the planned four campaigns at the :envihab facility in Germany, incorporating electrical muscle stimulation as a countermeasure. [Ed. Note: For more information regarding the :envihab facility, visit: https:// .] In addition, to this countermeasure, we are also assessing the potential of electrical impedance myography (EIM) and muscular ultrasound (MUS) to serve as surrogate measures of muscle health in comparison to standard strength and functional measures. These measures are being obtained in all individuals participating in the studies, including those undergoing no or different countermeasures than electrical muscle stimulation alone. At this point, we have begun analysis of these data sets, if nothing else than to help further refine our plans as soon as all the data is collected.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Abstracts for Journals and Proceedings	Macaulay TR, Rutkove S, Peters BT, Oddsson L, Wood SJ, Bent LR, Valero-Cuevas FJ, Lawrence EL, Strock N, Young M, Reschke M, Bloomberg JJ, Mortreux M, Bouxsein ML, Beheshti A, Abitante T, Newman D, Dunlap K, Sawyer A, Huang A, Kramer L, Laurie SS, Macias BR. "Countermeasures for mitigation of sensorimotor decrements following head-down bed rest." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025 Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025